

Experimental Investigation of Mechanical Properties of Hybrid Composites

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Abstract

With the increasing need for sustainable and eco-friendly materials, hybrid composites have gained attention as a potential solution due to their ability to combine different types of fibers with the best properties. In this study, we aim to investigate the mechanical properties of hybrid composites using banana fiber and flax fiber. By mixing different proportions of banana and flax fibers, we aim to evaluate the hardness, impact, tensile and flexural strengths of these composites. We can observe that the combination of both fibers will result in a synergistic effect, leading to increased mechanical properties. The optimal ratio of banana fiber to flax fiber is to vary depending on the intended application of the composite. Our findings will contribute to the development of more sustainable and high-performance materials for various industries, including construction and automotive. The potential benefits of using hybrid composites, especially those composed of eco-friendly and sustainable fibers, are numerous and could lead to a more sustainable future.

Keywords

Flax fiber, Banana fiber, Hybrid composite, Hand layup method, Mechanical properties, Tensile strength, Flexural strength, Impact strength

1. Introduction

Natural fiber reinforced composites (NFRCs) are a family of materials that have attracted a lot of attention recently because of their superior mechanical characteristics, low price, capacity to be recycled, and environmental friendliness. A polymer matrix and a natural fiber reinforcement make up the two separate parts of these materials. Jute, hemp, sisal, bamboo, kenaf, coir, flax, cotton, wool, and silk are just a few examples of the many plant and animal sources from which the natural fiber reinforcement can be generated. The polymer matrix, which is commonly made of petrochemicals, can either be thermosetting or thermoplastic.

NFRCs have a number of benefits over conventional composite materials that are only made of petrochemicals. First off, natural fibres are a sustainable and environmentally friendly option because they are a renewable resource that are simple to develop and harvest. Second, the finished

composite product may weigh less due to the fact that natural fibres are often lighter than conventional reinforcement materials like glass or carbon fibres. Thirdly, NFRCs are ideal for a variety of applications due to their superior mechanical qualities, which include high tensile strength, stiffness, and impact resistance.

The study of NFRCs has expanded significantly over the previous ten years, and numerous processing methods have been created to produce these materials. Compression moulding, injection moulding, extrusion, and hand layup are a few of the most popular methods. The choice of technique will rely on the particular requirements of the finished composite product because each of these methods has benefits and drawbacks of its own.

We will give an overview of the present status of research on NFRCs in this paper, along with a discussion of the many natural fiber types that are frequently utilized, the manufacturing processes that are employed to create these materials, and the mechanical characteristics of NFRCs. We will also go through some of the potential uses for NFRCs, such as their utilization in the construction, aerospace, and automotive sectors. Finally, we will discuss some of the issues that must be resolved in order to promote the creation and marketing of NFRCs.

2. Literature Review

Natural fiber reinforced composites have relatively minimal negative effects on the environment, making them ideal for a variety of sustainable engineering applications [1]. NFRC was found to perform better than synthetic composites since it is low density, inexpensive, and environmentally benign. Natural fiber composites are therefore highly advantageous for use in both engineering and commercial applications [2, 3]. Carbon fiber-based composite materials are a great example for being more affordable using recycling techniques [4]. Using a metal coating process with NFRPC could enhance the impact resistance and toughness properties of the fiber. The improvement of NFRC's impact toughness may be influenced by the thickness and size of stitching [5]. Natural fiber-reinforced hybrid composites have variations in mechanical properties with various volume fractions or weight fractions in composites of the same combination [6]. In comparison to untreated fiber composites, treated fiber-reinforced hybrid composites exhibit greater thermal stability. The hybrid composites with fiber reinforcement that had been treated had less kinetic and less water absorption. The fiber and matrix had greater adherence as a result [7, 8, 9]. Some studies have also proved combining a natural with a synthetic fiber over two natural fibers have shown improvement in their mechanical properties [10, 11]. In this experiment, among the three natural fiber composites (flax, hemp, and kenaf), flax fiber-reinforced composites with 00 fiber orientations exhibit the highest tensile strength value of 142.91 MPa. It is also discovered that changing the fiber orientation has an impact on the tensile strength values. [12]. According to experimental results, lengthening the fiber increases its tensile, flexural, and impact strengths [13]. It has been demonstrated that the mechanical properties of kenaf fiber reinforced epoxy composite, including flexural strength, impact strength, tensile strength and hardness, increase with fiber content. Kenaf fiber reinforced epoxy composites with a 35% fiber content can replace synthetics in a number of industries, including automotive, furniture, and packaging. [14]. According to the

testing findings, chemically processed hemp and epoxy composite parts will be a more desirable alternative to man-made or synthetic fiber composite parts in automotive and building applications [15]. The mechanical properties of composite materials are improved by chemically treating natural fibres to promote their adherence to the matrix. Natural fiber may replace synthetic fibres in many applications and is quickly becoming one of the sustainable and renewable resources in the composites industry [16]. D.K.K. Cavalcanti conducted an experiment showing untreated pure jute composites as compared to the strength of jute + sisal and jute + curaua composites increased by roughly 68% and 77%, respectively, while the strength of jute + glass fiber composites increased by roughly 90%. [17]. Similar experiment by F Sayer has shown the effect of varying resins. The best materials were found to be unmercerized cotton or bamboo-based fibres mixed with an epoxy or UV matrix to provide low visibility on all modalities and a flexural modulus at 71% that of carbon fiber. [18]. M. Arul has further verified the impact of resin used by combining coir fiber with hybrid composites made of 50% coir and 50% SLS fiber, improved tensile strength and flexural strength were found. According to the impact test, the specimen made with a polyester matrix but containing 50% coir and 50% SLS fiber has the highest impact strength [19]. Composites have emerged as a popular material for commercial aircraft, with the Airbus A380 using 25% of them, the Boeing 787 "Dreamliner" using 50%, and the A350 using 53%. The key to increasing fuel efficiency in composites is weight reduction [20]. The overall weight of the aero plane is reduced by 35% as a result of the development of natural fiber reinforced composites. The first NFRC for the inner door panel of flax and sisal hybrid epoxy matrix was used in the E-class Mercedes-Benz and demonstrated a notable weight reduction of roughly 20% [21]. Since, no one tested Flax and Banana Hybrid Composites, we think this might be useful in some of any applications

3. Materials and Methodologies

Banana fiber being durable and light in weight, flax fiber known for its ability to withstand high levels of tension, when combined has a probability of yielding a hybrid composite with higher mechanical properties.

3.1. Fibers Used

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3.1.1. Flax Fiber

Flax fiber, also known as linen fiber, is a natural, plant-based fiber that has been used for thousands of years in the production of textiles, clothing, and other household items. It is derived from the stems of the flax plant (*Linum usitatissimum*) and is one of the strongest and most durable natural fibers in the world.

3.1.2. Banana Fiber

Banana fiber is a natural, renewable, and eco-friendly fiber that is extracted from the stem of the banana plant. It is a strong, durable, and lightweight material that has been used for centuries in various cultures for making clothing, baskets, and other household items.

3.1.3. Epoxy Resin and HY95 hardener

The resin-hardener mixture is used to bind different fiber layer thicknesses. Under normal room temperature, the best binding property is provided by LY556 epoxy resin and HY951 hardener. According to studies, the ideal ratio for blending resin and hardener is 9:1, therefore we have used the same ratio.

3.2. Preparation of composites by hand layup method

A few primitive steps have been taken prior to the hand layup process, including the calculation of individual weights of single fiber layer of banana and flax. According to studies, the ideal ratio of fiber to resin with to weight was found out to be 1:3. Hence, after the weight analysis of individual fibers layers, resin in the required ratio was taken by weight. In our procedure we have considered an extra of 30gm, which is supposed to be the resin that would be squeezed out of the layers during the fabrication procedure. Also, based on the weights, we calculated the combination of flax and banana fiber layers to be present in a particular hybrid composite as shown in *table 4.1*.

- FB1 : 20%F + 80%B
 FB2 : 40%F + 60%B
 FB3 : 50%F + 50%B
 FB4 : 60%F + 40%B
 FB5 : 80%F + 20%B

Composition (B = Banana, F= Flax)	Flax fiber Layers (10 gms)	Flax Weight measured	Banana fiber layers (14 gms)	Banana weight measured	Total weight of fibers	Resin required (1:3)	Hardener required (9:1)
FB1	1	10	3	46		168	16.8
FB2	2	21	3	41	62	186	18.6
FB3	3	30	2	27	57	171	17.1
FB4	5	40	2	25	65	195	19.5
FB5	5	51	1	12	63	189	18.9

Table 4.1 Weight analysis

For the preparation of our hybrid fiber composite, we have used the hand layup method. At first, to remove any moisture, the fibres are first allowed to dry in the sunlight. Then, a wax polish, a releasing agent, was correctly spread out across a flat table for straightforward removal during post-processing. A full application of the LY556 epoxy resin and HY951 hardener mixture between two layers is required for optimal binding. A layer of banana or flax fiber is placed upon the above setup, as per the decided compositions. Application of the epoxy resin and hardener mixture follows the previous step. After the application of resin, a second layer of fiber is placed on top of the epoxy resin and hardener mixture layer. A roller is then used upon the fiber layers to roll out the air bubbles present in between them, also to make the binding better. Layers of epoxy resin and hardener mixture and fibers are then followed until the desired composition is achieved. The excess resin that we added in the primitive stage is now removed in the squeezing process. The excess resin was added to insure proper binding of the layers. A layer of laminate having the wax polish applied is placed on top of our fiber layers. Having placed a uniform load on top of it, it's left undisturbed for 24 hours.

4. Results & Discussions

To analyze the mechanical properties of the hybrid composite, we have subjected them to tensile test, flexural test, hardness test and impact test.

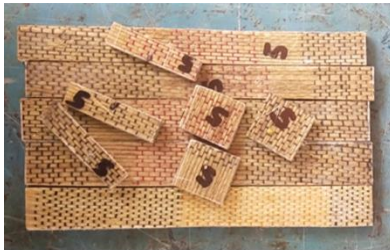


Fig 4.1 (a)



Fig 4.1 (b)

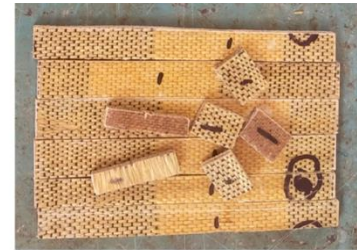


Fig 4.1 (c)

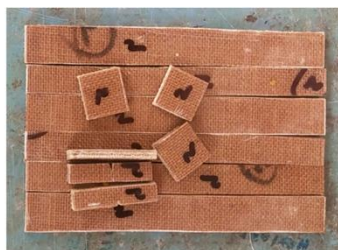


Fig 4.1 (d)



Fig 4.1 (e)

- Fig 5.1 (a) - **FB1** specimens for tensile, flexural, hardness and impact tests
- Fig 5.1 (b) - **FB2** specimens for tensile, flexural, hardness and impact tests
- Fig 5.1 (c) - **FB3** specimens for tensile, flexural, hardness and impact tests
- Fig 5.1 (d) - **FB4** Banana fiber specimens for tensile, flexural, hardness and impact tests
- Fig 5.1 (e) - **FB5** Banana fiber specimens for tensile, flexural, hardness and impact tests

4.1. Tensile Test

The tensile test is performed on a rectangular specimens with dimension 200x25x6 as shown in *fig 4.3*. The specimen is tested using a UTM (universal testing machine). The procedure begins with marking 50mm from the either ends of the specimen. Using the flat jaws of the UTM, the specimen is then fixed by the 50mm mark made. The specimen is then subjected to tensile loads which are gradually increased. The load at which the specimen breaks is recorded as the breaking load of that specimen as shown in *table 4.1*. The test is repeated for three times each for the 5 different compositions of hybrid fiber made.



Fig 4.2 UTM

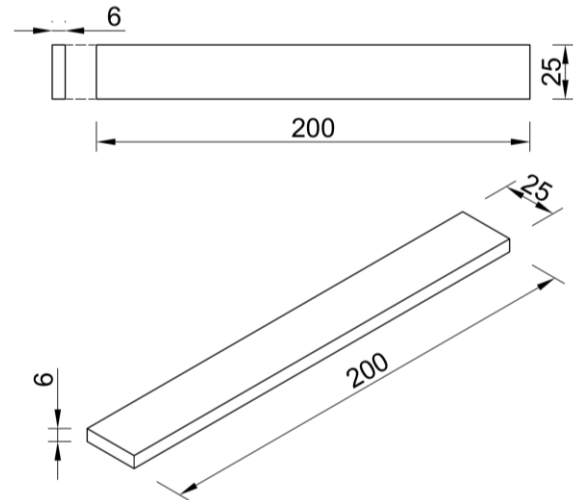


Fig 4.3 Tensile test specimen

Tensile strength is calculated by the formula,

$$\text{Tensile Strength, } \sigma = \frac{F}{A} \dots\dots\dots (i)$$

Where, A = 0.025x0.006 m²

Composition	No. of Divisions (d)	Elongation (mm)	Breaking Load (F = 0.1d kN)	Tensile Strength ($\sigma = F/A$ Pa)	Average Tensile Strength (σ_a) (Mpa)
FB1	4	6	0.4 kN	2.67 MPa	3.04 MPa
	5.2	10	0.52 kN	3.47 MPa	
	4.5	8	0.45 kN	3 MPa	
FB2	5.2	4	0.52 kN	3.47 MPa	3.13 MPa
	4.2	3	0.42 kN	2.8 MPa	
	4.7	4	0.47 kN	3.13 MPa	

FB3	4.6	6	0.46 kN	3.07 MPa	2.95 MPa
	4.3	6	0.43 kN	2.87 MPa	
	4.4	6	0.44 kN	2.93 MPa	
FB4	5	2	0.5 kN	3.33 MPa	3.33 MPa
	5	5	0.5 kN	3.33 MPa	
	5	3	0.5 kN	3.33 MPa	
FB5	4.6	2	0.46 kN	3.07 MPa	2.67 MPa
	3.4	6	0.34 kN	2.27 MPa	
	4	4	0.4 kN	2.67 MPa	

Table 4.1 Tensile test

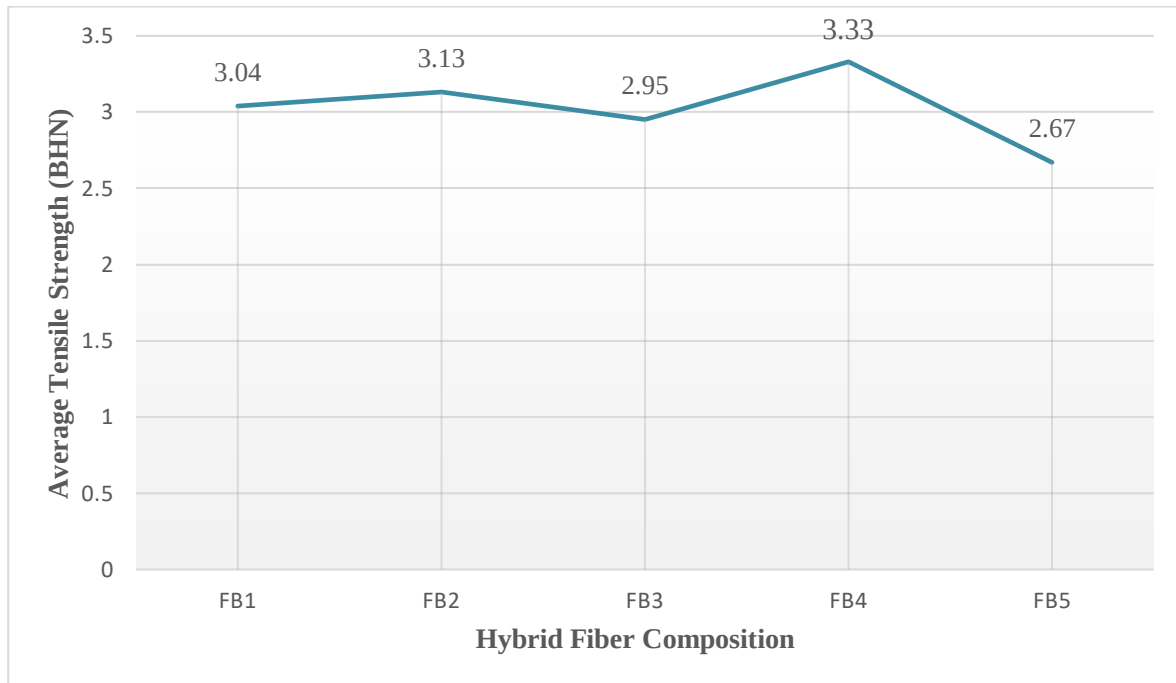


Fig 4.4 Tensile Strength for different compositions

From the *table 4.1* and the graph above we can observe the hybrid fiber having FB4 has shown the highest resistance against the tensile loads applied with its tensile strength as 3.33 Mpa.

4.2. Flexural Test

Similar to tensile test the UTM is used to perform the flexural test. Here, the specimen breadth changes to that of tensile test, i.e., 200x20x6 as shown in *fig 4.6*. Here the making is done at the center along the length. This point is where the load would be applied through the UTM. The readings are noted down in a tabular form during the test as in *table 4.2*. The test is repeated for three times each for the 5 different compositions of hybrid fiber made.



Fig 4.5 Specimen undergoing flexural test

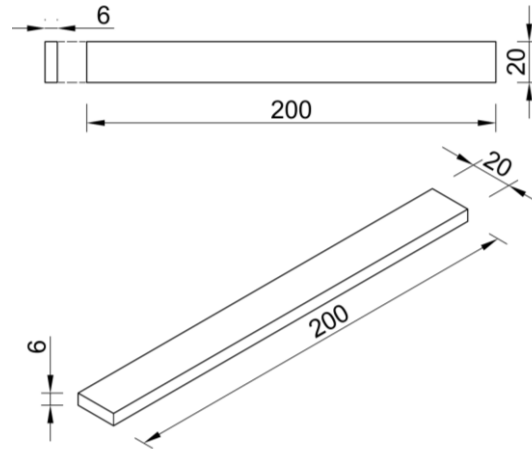


Fig 4.6 Tensile test specimen

Flexural strength is calculated by the formula,

$$\text{Flexural Strength, } \sigma = \frac{3Fl}{2bd^2} \dots\dots\dots (ii)$$

Where, $l = 0.2\text{m}$

$b = 0.02\text{m}$

$d = 0.006\text{m}$

Composition	No. of Divisions (d)	Breaking Load (F = 0.1d kN)	Flexural Strength ($\sigma_a = 3Fl/2bd^2$)	Average Breaking Load (F _a) (N)	Average Flexural Strength (σ_a) (Mpa)
FB1	7.5	0.75 kN	312.5 MPa	775	322.22 MPa
	8	0.8 kN	333.33 MPa		
	7.7	0.77 kN	320.83 MPa		

FB2	7	0.7 kN	291.67 MPa	750	312.5 MPa
	8	0.8 kN	333.33 MPa		
	7.5	0.75 kN	312.5 MPa		
FB3	7	0.7 kN	291.76 MPa	650	270.86 MPa
	6	0.6 kN	250 MPa		
	6.5	0.65 kN	270.83 MPa		
FB4	7	0.7 kN	291.67 MPa	723	301.39 MPa
	7.5	0.75 kN	312.5 MPa		
	7.2	0.72 kN	300 MPa		
FB5	7	0.7 kN	291.67 MPa	700	291.67 MPa
	7	0.7 kN	291.67 MPa		
	7	0.7 kN	291.67 MPa		

Table 4.2 Flexural test

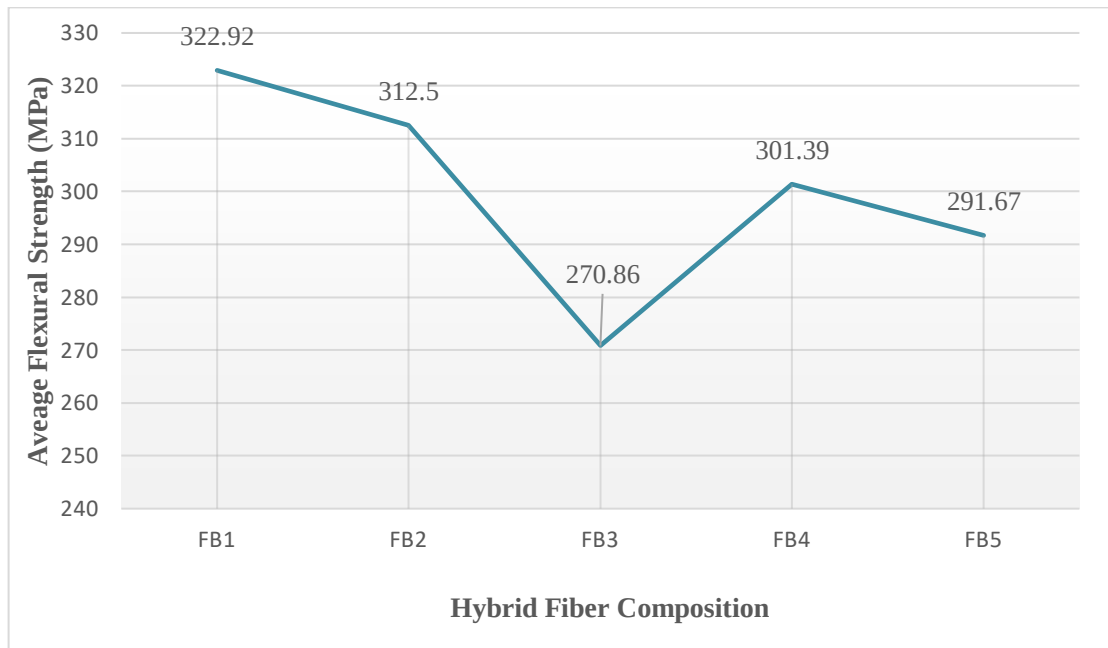


Fig 4.7 Flexural Strength for different compositions

From the *table 4.2* and the graph above we can observe the hybrid fiber having FB1 banana fiber has shown the highest resistance against the flexural loads applied, with flexural strength as 322.92 MPa.

4.3. Hardness Test

For hardness test, Brinell hardness test has been preferred. The specimen for hardness test is shown in *fig 4.9* which is 30x30x6. Brinell hardness tester (*fig 4.8*) is used to perform the hardness test upon the hybrid composition. It is performed to understand the amount of resistance to permanent deformation the hybrid fiber has when, penetration of a harder material. The test is repeated for three times each for the 5 different compositions of hybrid fiber made and the values are noted in *table 4.3*.



Fig 4.8 Brinell hardness tester

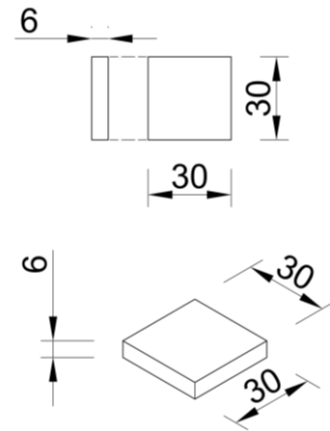


Fig 4.9 Hardness test specimen

Brinell hardness number is calculated by the formula,

$$\text{Brinell hardness number, BHN} = \frac{2P}{\pi D \cdot (D - \sqrt{D^2 - d^2})} \dots\dots\dots \text{(iii)}$$

Where, P = Load applied

D = diameter of ball indenter = 3.125 mm

d = diameter of indentation (mm)

Composition (B = Banana, F= Flax)	Load Applied, P (Kgf)	Indentation (mm)	Brinell Hardness Number (BHN)	Average Brinell Hardness Number (BHN)
FB1	100	1.9	31.65	34.36
	100	1.8	36.72	
	100	1.8	36.72	
FB2	100	1.5	53.14	53.14
	100	1.5	53.14	
	100	1.5	53.14	

FB3	100	2	28.15	28.15
	100	2	28.15	
	100	2	28.15	
FB4	100	2.4	18.14	17.53
	100	2.4	18.14	
	100	2.5	16.30	
FB5	100	2.1	25.13	25.13
	100	2.1	25.13	
	100	2.1	25.13	

Table 4.3 Hardness test

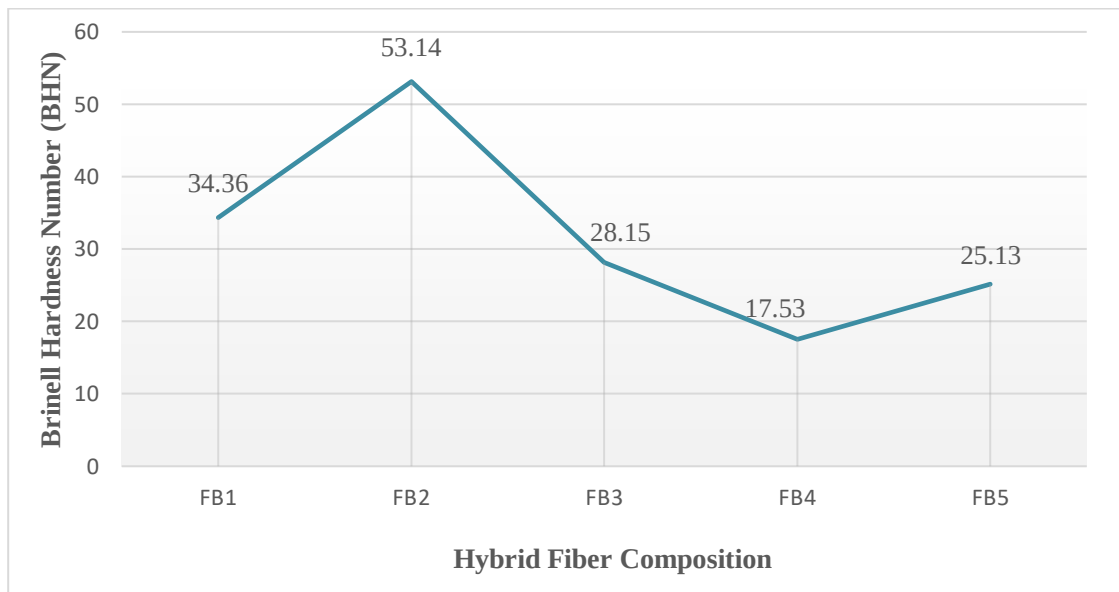


Fig 4.10 Hardness for different compositions

From the *table 4.3* and the graph above we can observe the hybrid fiber having 40% flax fiber and 60% banana Fiber has shown the highest resistance to permanent deformation against any penetration due to a harder substance. The Brinell hardness number of this hybrid fiber composite is observed to be 53.14 BHN.

4.4. Impact Test

The specimen for impact test is shown in *fig 4.12* which is 55x13x6. Charpy impact tester is used to perform the impact test on the specimens of the hybrid fiber. The hybrid fiber composite might be subjected to shocks and impacts in practical use so this test is performed to determine its ability to withstand them. The test is repeated for three times each for the 5 different compositions of hybrid fiber made and the values are noted in *table 4.4*.



Fig 4.11 Charpy impact test on specimen

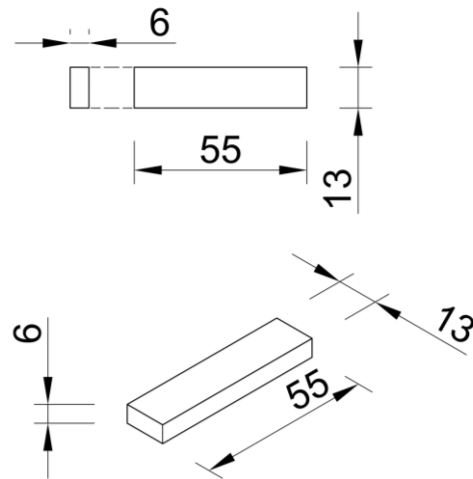


Fig 4.12 Impact test specimen

The results values from the four tests performed have been noted in a tabular format and have been studied to find out the best composition of the hybrid fiber for a particular mechanical property i.e., tensile strength, flexural strength, hardness or impact strength.

Impact strength is calculated by the formula,

$$\text{Impact Strength, } \sigma = \frac{E}{t} \dots\dots\dots (\text{iv})$$

Where, t = thickness of specimen, 6mm

Composition	No. of Divisions (d)	Impact Energy (2d) (J)	Impact Strength (E/t) (t=0.006m)	Average Impact Strength (J)
FB1	0.5	1 J	166.67	388.3
	1.5	3 J	500	
	1.5	3 J	500	

FB2	1.5	3 J	500	721.6
	4	8 J	1333.33	
	1	2 J	333.33	
FB3	1	2 J	333.33	333.33
	1	2 J	333.33	
	1	2 J	333.33	
FB4	1	2 J	333.33	666.67
	4	8 J	1333.33	
	1	2 J	333.33	
FB5	2	4 J	666.67	500
	1	2 J	333.33	
	1.5	3 J	500	

Table 4.4 Impact test

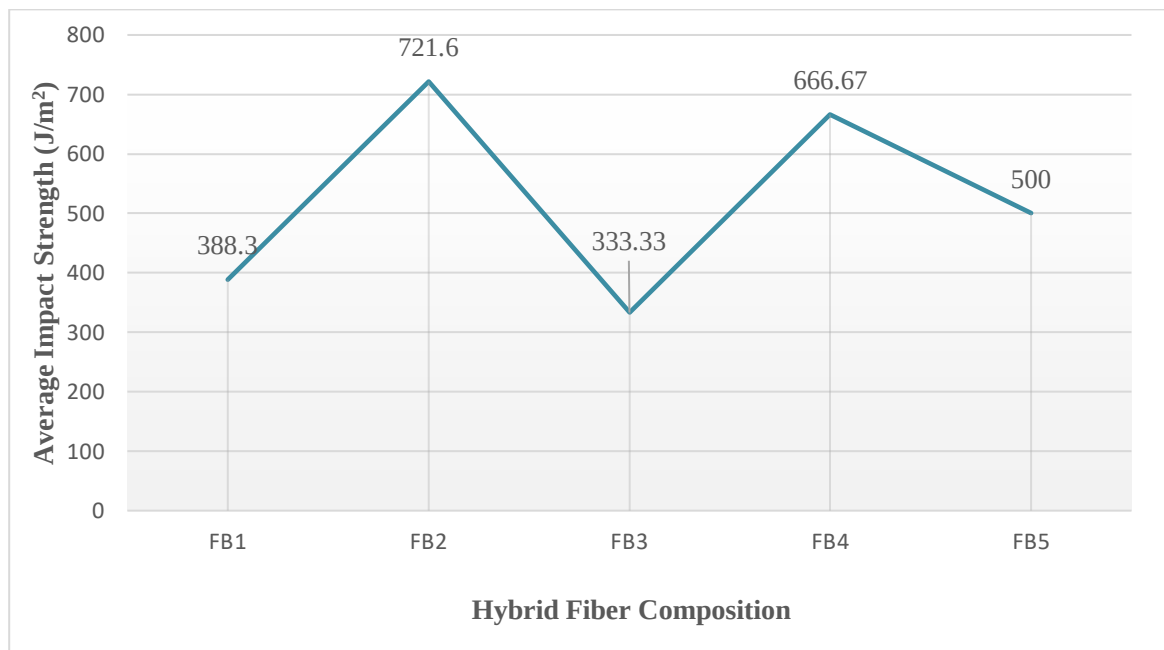


Fig 4.13 Impact Strength for different compositions

From the *table 6.2* and the graph above we can observe the hybrid fiber having FB2 fiber has shown the highest impact strength as 712.6 J/m².

5. Conclusion and Future Scope

From the above performed tests,

FB1 has the best flexural properties among the 5 compositions and having good tensile properties. Whereas, it lacks in impact resistance and hardness properties.

FB2 has the best impact resistance and hardness properties among the 5 compositions and having good tensile and flexural properties but having the least hardness property.

FB4 has the best tensile, properties among the 5 compositions and having good hardness and impact resistance properties. Whereas, it lacks in flexural properties by a small difference from the highest flexural strength value.

Composition	Tensile Strength	Flexural Strength	Hardness	Impact Strength
FB1	3.04 MPa	<u>322.92 MPa</u>	34.36	388.3
FB2	3.13 Mpa	312.5 MPa	<u>53.14</u>	<u>721.6</u>
FB3	2.95 Mpa	270.86 MPa	28.15	333.33
FB4	<u>3.33 Mpa</u>	301.39 MPa	17.53	666.67
FB5	2.67 MPa	291.67 MPa	25.13	500

Table 6.5 Experimental sum-up

** Highest value obtained in the respective tests*

We can conclude stating the best combination of hybrid fiber composite for different properties required. Also, from the results above we can observe that the FB2 has the best impact resistance and hardness properties. It performed the best while considering all the tests performed. It may be considered as a composition which can be used where the use of all four mechanical properties studied in this experiment are required.

The attempt to analyze the mechanical properties of flax and banana hybrid fiber composite haven't been made yet. This paper can be referred to while writing, experimenting on the hybrid fiber composite of flax fiber and banana fiber.

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